

2026-06-02 - BLE Peripheral Advertising

Goal

Set up the nRF52840 as a Bluetooth Low Energy (BLE) peripheral that advertises a device name and a custom manufacturer data payload, connectable by a phone or PC.

What you will learn

- How to enable and initialize the Zephyr BLE stack
- How to construct advertising and scan response data
- How to start and stop advertising
- How to handle connect and disconnect events
- How to read the device address and advertised name with a phone

Overview

BLE advertising is the mechanism by which a peripheral announces its presence. The nRF52840 has a built-in radio that Zephyr drives through the `bt_` API. No external hardware is required — just the SoC and a power supply.

prj.conf

```
CONFIG_BT=y
CONFIG_BT_PERIPHERAL=y
CONFIG_BT_DEVICE_NAME="MyZephyrDevice"
CONFIG_BT_DEVICE_APPEARANCE=0
CONFIG_LOG=y
```

Advertising data structure

```
static const struct bt_data ad[] = {
    BT_DATA_BYTES(BT_DATA_FLAGS, BT_LE_AD_GENERAL | BT_LE_AD_NO_BREDR),
    BT_DATA(BT_DATA_NAME_COMPLETE, CONFIG_BT_DEVICE_NAME,
            sizeof(CONFIG_BT_DEVICE_NAME) - 1),
};

/* Manufacturer-specific data in scan response */
static const uint8_t manuf_data[] = { 0xFF, 0x07, 0x01, 0x02, 0x03 };
static const struct bt_data sd[] = {
```

```
BT_DATA(BT_DATA_MANUFACTURER_DATA, manuf_data, sizeof(manuf_data)),  
};
```

Starting advertising

```
bt_enable(NULL);    /* initialise the BLE stack */  
  
bt_le_adv_start(BT_LE_ADV_CONN, ad, ARRAY_SIZE(ad), sd, ARRAY_SIZE(sd));  
  
BT_LE_ADV_CONN = connectable undirected advertising at 100 ms interval.
```

Connection callbacks

```
static void connected(struct bt_conn *conn, uint8_t err) {  
    printk("Connected\n");  
}  
static void disconnected(struct bt_conn *conn, uint8_t reason) {  
    printk("Disconnected (reason %u)\n", reason);  
    bt_le_adv_start(BT_LE_ADV_CONN, ad, ARRAY_SIZE(ad), sd, ARRAY_SIZE(sd))  
}  
  
BT_CONN_CB_DEFINE(conn_callbacks) = {  
    .connected      = connected,  
    .disconnected   = disconnected,  
};
```

Verifying on a phone

- **Android:** nRF Connect app → Scanner tab → find "MyZephyrDevice"
- **iOS:** LightBlue app → Peripherals tab → find "MyZephyrDevice"
- **Linux:** bluetoothctl scan on → look for the device name

Why this matters

BLE advertising is the entry point to all BLE applications — sensor beacons, wearables, and IoT devices all start here. Once advertising works, you can add GATT services to expose data to connected clients.

Practice tasks

1. Flash and scan with nRF Connect — confirm the device name appears.
2. Change CONFIG_BT_DEVICE_NAME and rebuild — confirm the new name.
3. Add a second byte to manuf_data containing a counter that increments each time the device boots (combine with NVS from Day 22).

Example folder

See `src/2026-06-02_BLE_Peripheral_Advertising` for the day's code.

Next topic

Tomorrow we look at SPI communication to interface with external peripherals.